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Key Points:
- A key Late Ordovician paleomagnetic pole for the Yangtze Block is reported
- The Yangtze Block was at tropic latitudes during the Late Ordovician
- The data favor South China' proximity to Australia in the Late Ordovician

Supporting Information:
- Tables S1 and S2

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New paleomagnetic results from Late Ordovician rocks of the Yangtze Block, South China, and their paleogeographic implications

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Abstract The paleogeographic relationship between South China and Australia during the Ordovician is important for understanding the configuration of South China in Gondwana. However, high-quality Ordovician paleomagnetic results for the Yangtze Block are scarce. Here we report the results of a new paleomagnetic study of the Late Ordovician limestones of Wangcang County in the northern Yangtze Block, performed in order to constrain the paleoposition of South China. Two magnetic components were isolated by detailed stepwise thermal demagnetization. The low-temperature component falls close to the local current Earth's field direction. The site-mean direction obtained from the high-temperature component (HTC) carried by magnetite is $D/I = 132.6°/-35.2°$ ($\alpha_{95} = 3.6°$) after bedding correction, yielding a paleomagnetic pole at 45.8°S, 191.3°E (dp = 2.4°, dm = 4.2°). The HTC direction passed reversal and fold tests, and its corresponding pole differs from the available paleomagnetic poles since the Silurian of the South China Block. These results suggest that the remanent magnetization was probably acquired during the earliest stage of sedimentation. The high-temperature component yields a paleolatitude of 19.5°S, implying that the Yangtze Block was at tropic latitudes during the Late Ordovician. These new and reliable paleomagnetic results bridge the Ordovician data gap and favor the proximity between South China and Australia during the Late Ordovician.

1. Introduction

The paleogeographic relationship between South China and Australia in the Gondwanan frame is important for understanding both the position of South China in the Rodinia configuration and the dispersion of Gondwana and the accretion of the Asian continent. There is a general consensus that the South China Block (SCB) had a strong faunal affinity with eastern Gondwana during the early Paleozoic [*Burrett et al.*, 1990]. However, there is significant debate on the paleoposition of the SCB in Gondwana, in terms of whether it occupied a position on the margin of NW Australia in the Neoproterozoic [*Zhao and Cawood*, 1999; *Zhou et al.*, 2002] or from the Neoproterozoic to the early Paleozoic [*Yang et al.*, 2004; *Cawood et al.*, 2013], or whether it migrated from the center of Rodinia to a position external to Gondwana during the Neoproterozoic to the early Paleozoic [*Li et al.*, 2008, 2013; *Zhang et al.*, 2013]. Two extreme reconstruction models have been proposed. *Yang et al.* [2004] suggested that the SCB abutted northwestern Australia and both blocks maintained a relatively constant position in the early Paleozoic. In contrast, *Li et al.* [2008] argued that the SCB broke away from the middle of Australia and Laurentia due to the occurrence of a super mantle plume from 820–750 Ma and then migrated from northeast Australia to northwestern Australia during the late Proterozoic to early Paleozoic.

The main reason for the uncertain paleoposition of South China in the Gondwanan frame is the lack of a high-quality paleomagnetic data set for the early Paleozoic. Apart from widespread late Mesozoic remagnetization related to the intense tectonic magmatism in the Yangtze Block [*Kent et al.*, 1987; *Wang and Van der Voo*, 1993; *Bai et al.*, 1998b], Ordovician paleomagnetic data were reported [*Fang et al.*, 1990; *Wu et al.*, 1999] in which the reliability quality index *Q* [*Van der Voo*, 1990] is generally lower than 5. These results are controversial. *Fang et al.* [1990] obtained a high paleolatitude of 48°S in the Early Ordovician for the Yangtze Block, while *Wu et al.* [1999] argued that the paleolatitude was 9.5°S in Early and Middle Ordovician time. Although high-quality paleomagnetic results were obtained for the Middle Cambrian [*Bai et al.*, 1998a; *Yang et al.*, 2004] and Silurian [*Opdyke et al.*, 1987; *Huang et al.*, 2000], the lack of reliable Ordovician

paleomagnetic data results in a large gap of about 80 Ma from the Middle Cambrian to the Silurian, which hampers the establishment of an integrated apparent polar wander path (APWP) for the Yangtze Block in the early Paleozoic. Therefore, high-quality Ordovician paleomagnetic data from the Yangtze Block are important for verifying the early Paleozoic paleogeographic reconstruction for South China and its proximity with Australia. Resolving the paleogeographic relationship between South China and Australia during the early Paleozoic may also improve our understanding of South China in the Rodinia configuration.

In this paper, we report new Late Ordovician paleomagnetic results from the Wangcang area in the northern Yangtze Block in order to provide a key paleopole for the definition of an integrated Paleozoic APWP for the Yangtze Block and to further constrain the tectonic relationship between South China and East Gondwana.

2. Geological Setting and Sampling

Our sampling areas in Wangcang county (32.4°N, 106.3°E), in Sichuan province, are located in the northern Yangtze Block (Figure 1a). The Upper Ordovician Pagoda Formation is a neritic facies and is widespread in the study areas. The lithology is relatively straightforward and fossils are abundant. The Pagoda Formation is 12–40 m in thickness and consists of gray, purplish red medium- to thick-bedded limestones characterized by the shrinkage cracks and is easy to recognize due to the occurrence of abundant nautiloids and the large, pagoda-like giant Orthocones. *Sinoceras chinense* sp. are the dominant and conspicuous fossils of the formation. Trilobites are also abundant in the Pagoda limestones, e.g., *Hammatocnemis* sp., *Nankinolithus* sp., *Asaphus* sp., *Remopleurides* sp., *Birmanites* sp., and *Illaenus* sp.. Other fossil groups include *Sowerbyella* sp., *Arcuaria* sp., and *Trochonema* sp.. Based on these abundant fossils, the formation is well dated and its age is assigned to the middle Sandbian to early Katian, namely, the middle to late Caradoc [*Zhou et al*., 2000; *Zhan and Jin*, 2007]. A multiphase folding occurred during the Jurassic-Cretaceous from the influence of the Yanshan movement of southern Qinling [*Wei et al*., 1997; *Pei et al*., 2009; *Sun*, 2011], leading to the development of box-shaped, short-axis folds with relatively few faults (Figure 1b).

A total of 479 samples from 45 sites were collected from the Upper Ordovician Pagoda Formation in the northern Wangcang area, using a portable drill and oriented with a magnetic compass. According to the distribution of fresh outcrops, 31 sites are distributed on both northern and southern flanks of the folds (Figure 1b), which enables the application of fold tests. Each site consists of 10–13 samples across 0.9–2.5 m. In addition, three short sections of more than 10 m were sampled continuously in order to define the potential magnetic polarity sequence. In the laboratory, each sample was cut into one or two 23 mm long, 25 mm diameter specimens with a diamond saw. All 479 specimens were subjected to stepwise thermal demagnetization, and 32 specimens were used for rock magnetic tests in order to characterize the magnetic mineralogy.

3. Laboratory Techniques

Specimens were subjected to progressive thermal demagnetization in 14–18 steps, using an ASC-TD48 oven with an internal residual field lower than 10 nT. Natural remanent magnetization (NRM) was measured using a 2G-755 magnetometer housed in a magnetically shielded room which reduced the ambient geomagnetic field to around 300 nT. The demagnetization temperature interval was 30–60°C below 450°C and 10–20°C from 450–580°C. Demagnetization results were analyzed using principal component analysis [*Kirschvink*, 1980] and Fisher statistics [*Fisher*, 1953], or using the intersection of sector-constrained great circles combined with directly observed directions [*McFadden and McElhinny*, 1988].

Isothermal remanent magnetization (IRM) was imparted and saturation IRM was demagnetized by applying reversed fields using an ASC IM-10-30 impulse magnetometer. The remanences were measured using a JR-6A spinner magnetometer. To characterize the magnetic carriers, IRM in fields of 2.4 T, 0.4 T, and 0.12 T were imparted along the *Z*, *Y*, and *X* axes of the samples, respectively, and were then thermally demagnetized [*Lowrie*, 1990]. An ASC-TD48 oven was used for demagnetization and a JR-6A spinner magnetometer for the measurements. All experiments were carried out at the Key Laboratory of Paleomagnetism and Tectonic Reconstruction of the Ministry of Land and Resource, Institute of Geomechanics, Beijing.

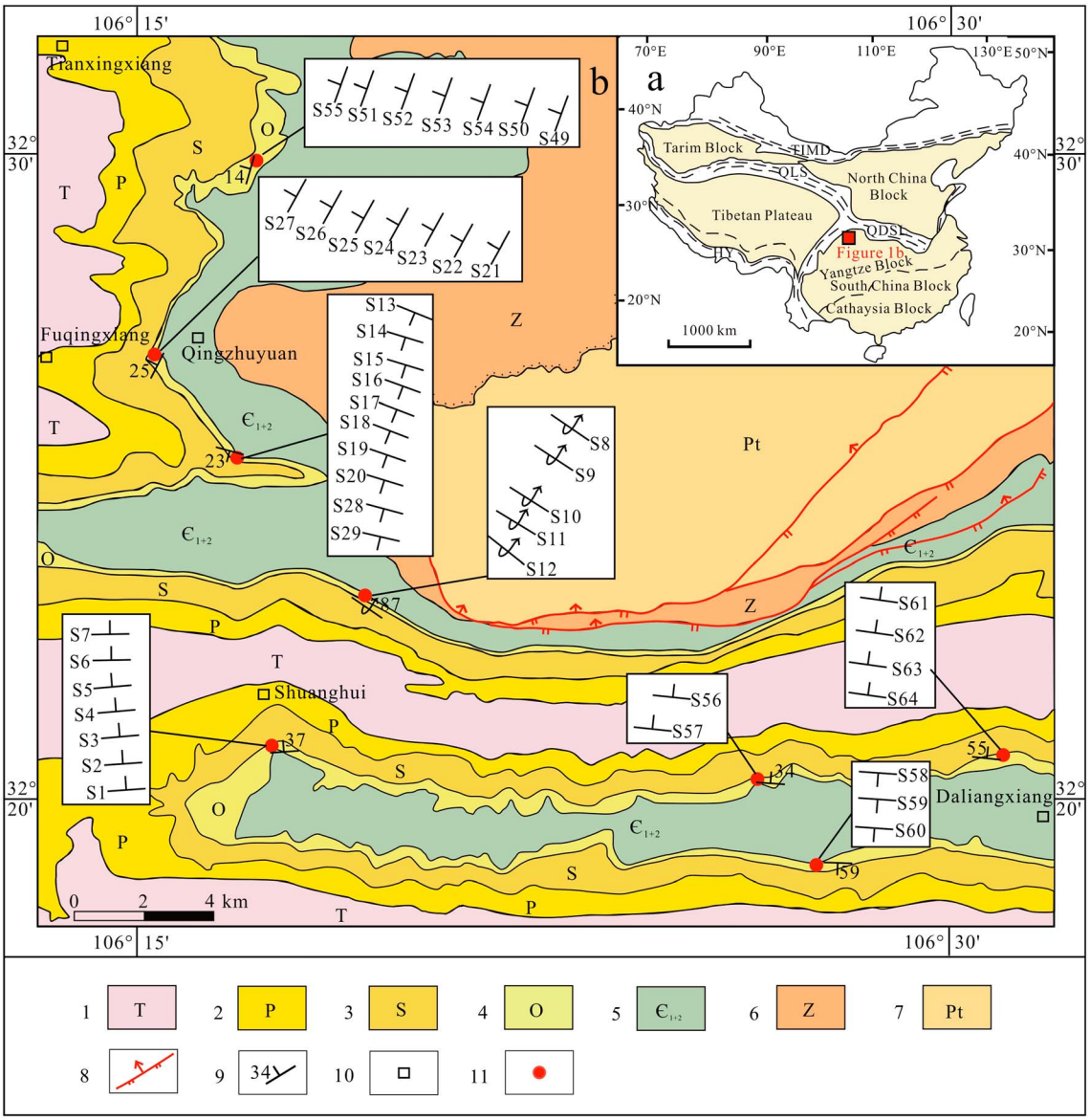


Figure 1. (a) Simplified geotectonic framework of China showing the location of the sampling area (modified from *Wang and Zhou* [2012]). TIMD = Tianshan-Inner Mongolia-Daxinganling fold belt, QLS = Qilianshan fold belt, QDSL = Qinling-Dabie-Su-Lu orogenic belt, HY = Himalaya fold belt. (b) Geological sketch map of the sampling area in northern Wangcang county (32.4°N, 106.3°E), northern Yangtze Block. The strike direction for each sampling site is shown in the insets. 1, Triassic; 2, Permian; 3, Silurian; 4, Ordovician; 5, Lower-Middle Cambrian; 6, Sinian; 7, Proterozoic; 8, reverse fault; 9, attitude with dip angle of the bedding plane; 10, locality; and 11, sampling sites.

4. Results

4.1. Rock Magnetic Results

IRM acquisition and thermal demagnetization of a three-component IRM are considered to be powerful analytical techniques for determining the intrinsic coercivity and unblocking temperature spectra of magnetic minerals [*Lowrie*, 1990]. For the limestones of the Upper Ordovician Pagoda Formation in the sampling area, the IRM acquisition and demagnetization curves, and the response of samples to three-component IRM thermal demagnetization, can be used to define three categories of behavior. In the first category, the IRM acquisition curves initially rise steeply but do not reach saturation until 1.9–2.2 T. The backfield demagnetization curves exhibit an inflection point at about 60 mT, and the remanent coercivity is at least 200 mT, indicating the presence of both low- and high-coercivity magnetic minerals. The thermal demagnetization of the three-component IRM exhibits a monotonic decay to 580°C in the low-coercivity (0.4 T) curve and 680°C in

the high-coercivity (2.4 T) curve (Figures 2a and 2b), suggesting domination of the magnetic properties by magnetite and hematite; however, the proportion of these minerals varies within the rocks. The aforementioned magnetic behavior is typical, in particular, of the purplish red limestone samples.

In the second category, the IRM acquisition curves rise steeply below 200 mT and then reach saturation at about 1.8 T. The backfield IRM demagnetization curves define a remanent coercivity of about 100 mT (Figure 2c). In the case of the samples shown in Figure 2d, the IRM acquisition curve initially rises very steeply and reaches over 90% of the saturation IRM at 0.4 T. However, the curve continues to rise gently and does not saturate until 0.9 T, reflecting the dominance of low-coercivity minerals. Thus, the difference between the IRM acquisition curves shown in Figures 2c and 2d is due to differences in the concentration of high-coercivity minerals. The three-component IRM thermal demagnetization curves reflect the predominance of low-coercivity minerals with two different unblocking temperature spectra, at around 320°C and 580°C, indicating the presence of pyrrhotite and magnetite, respectively (Figures 2c and 2d) [*Lowrie*, 1990].

In the third category, the IRM acquisition curve initially rises very rapidly and reaches over 90% of the saturation IRM at 0.4 T, indicating the dominance of low-coercivity minerals. The IRM backfield demagnetization curve defines a remanent coercivity of 100 mT. The three-component IRM thermal demagnetization curves reveal the presence of magnetic minerals with unblocking temperatures of 320°C, 350°C, and 560°C (Figure 2e), possibly reflecting an assemblage of pyrrhotite, titanomagnetite or maghemite, and magnetite.

4.2. Thermal Demagnetization Results

Samples from 45 sites from the Upper Ordovician Pagoda formation limestone were thermally demagnetized, defining two magnetic components. A low-temperature component is generally defined below 240°C–300°C, and a high-temperature component is defined above 480°C–500°C. The thermal demagnetization trajectories between 300°C and 480°C in the Zijderveld diagrams reveal an arched component, which is probably a resultant of the partially overlapping spectra of the low- and high-temperature components. Representative thermal demagnetization curves are shown in Figure 3, including those of normal and reversed polarities. The thermal demagnetization behavior from the NRM to 580°C indicates that the high-temperature component is mainly carried by magnetite, although three magnetic mineral combinations exist in the specimens. A minor hematite component is present in some of the purplish red specimens; however, it does not make a significant contribution to the natural remanent magnetization.

In some specimens, the thermal demagnetization behavior reveals that remanent directions generally become erratic at a threshold of 460°C–520°C, making it difficult to define linear segments at higher temperatures using principal component analysis. The magnetic minerals in these samples are magnetite and hematite (Figures 2a and 2b) and/or pyrrhotite and magnetite (Figures 2c and 2d), which may have resulted in magnetic mineral transformation during heating. In some cases, the magnetization intensity of rocks became weak above 460°C–520°C. However, the change of demagnetized directions is along a great circle (Figures 4a–4f), and therefore, the site-mean direction of the high-temperature component can be calculated using the intersection of sector-constrained great circles together with direct determination of remanent directions [*McFadden and McElhinny*, 1988]. Two examples of site-mean directions with normal and reversed polarities are shown in Figures 4g and 4h.

The high-temperature component cannot be isolated in nearly 60% of the total specimens, and in these cases the thermal demagnetization behavior is as follows: (1) a low-temperature component is isolated below 200°C or 300°C and directions become erratic above 200°C or 300°C, which may be related to magnetic mineral transformations during the heating process (Figures 5a–5c); (2) the direction is defined as a single component close to the present geomagnetic field (Figure 5d); (3) the directions can be isolated from a low-temperature component below 300°C and a component directed to the NNE from 300°C to 450°C that generally trends toward the origin with a short trajectory (Figure 5e); and (4) the demagnetization directions show more or less arced trajectories below 360°C, which is an indication of the overlapping of the component spectra (Figure 5f). In these samples, the high-temperature component is generally absent, and the magnetic mineral assemblage includes dominant pyrrhotite, titanomagnetite and/or maghemite, and minor magnetite (Figures 2d and 2e).

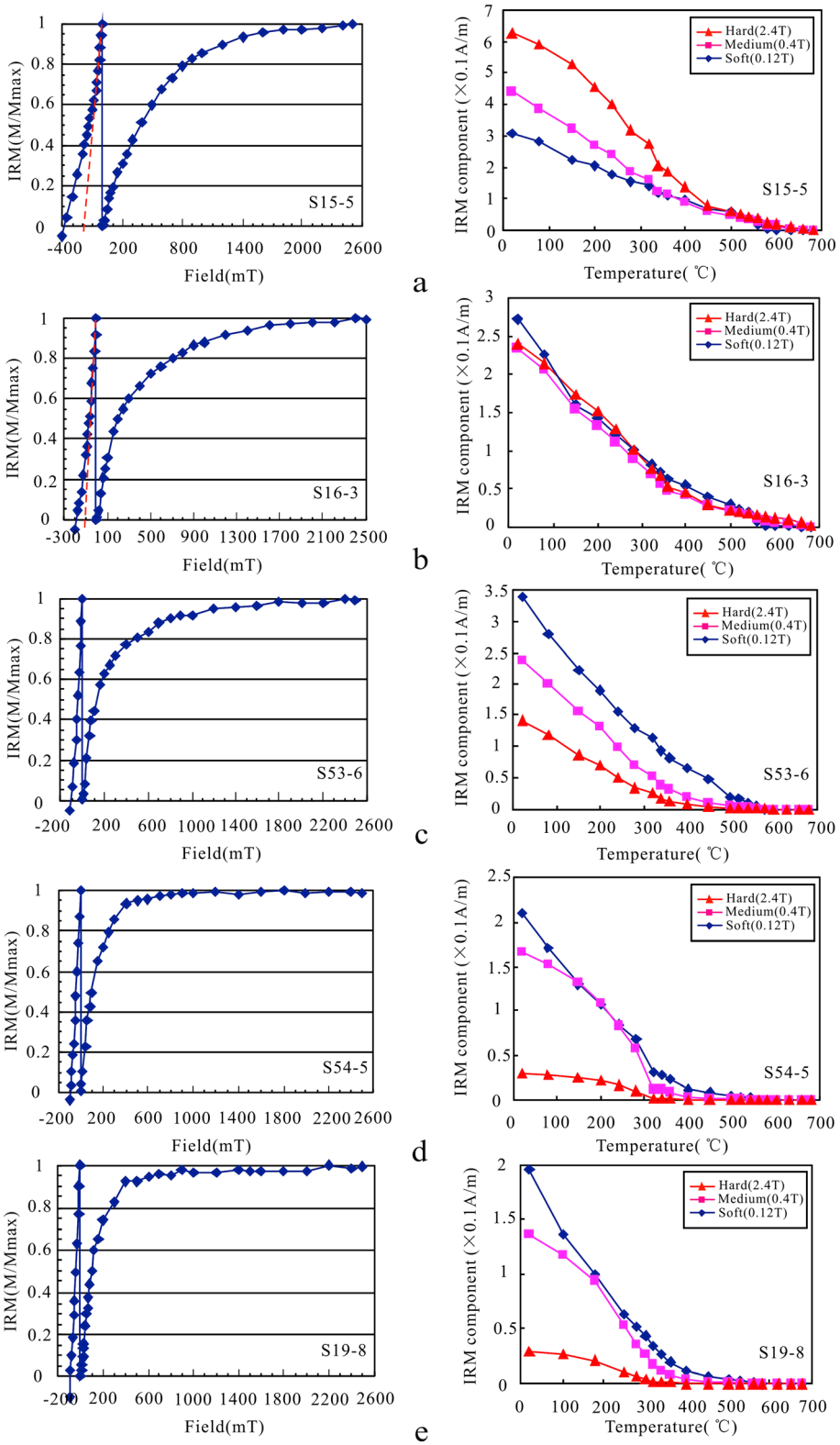


Figure 2. (a–e) Normalized isothermal remanent magnetization (IRM) acquisition, backfield IRM demagnetization, and three-component IRM thermal demagnetization curves for representative limestone specimens. Diamonds indicate the soft component (0.12 T field); squares indicate the medium component (0.4 T field); and triangles indicate the hard component (2.4 T field). The red broken line denotes the backfield demagnetization, and the remanent coercivity about 60 mT is indicated by the inflection point.

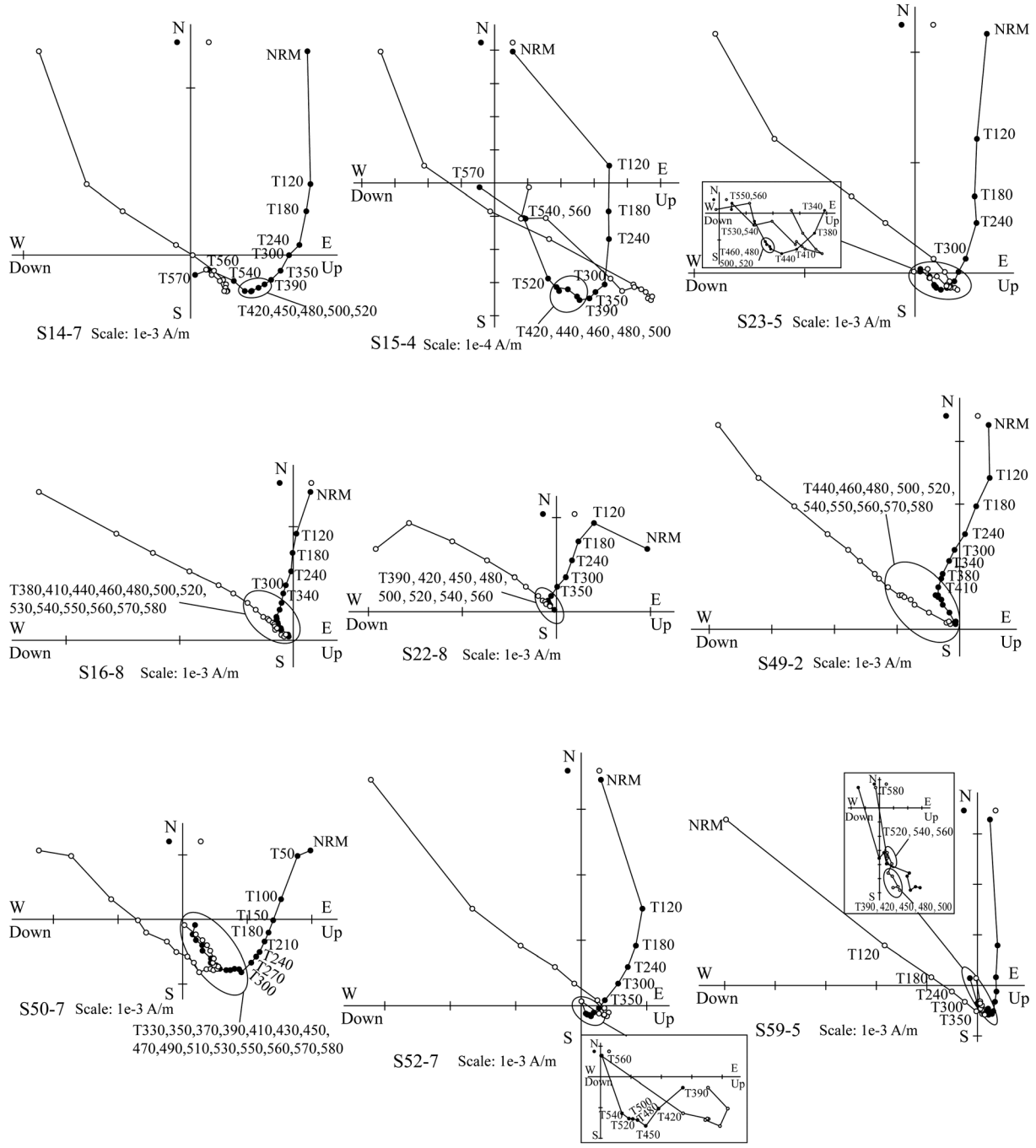


Figure 3. Orthogonal vector diagrams illustrating the response of representative specimens from limestones of the Upper Ordovician Pagoda Formation to thermal demagnetization (in situ coordinates). Solid (open) symbols denote the projection on the horizontal (vertical) planes. Key: T200 = 200°C. Low- and high-temperature components are clearly revealed.

The site-mean direction of the low-temperature component in in situ coordinates is Dg/Ig = 3.9°/54.0° (k = 362.8, α_{95} = 1.1°, and N = 45), and Ds/Is = 338.1°/53.1° (k = 4.0, α_{95} = 12.3°) after tilt correction (Figure 6). The ratio of precision parameters is maximum before unfolding (kg/ks = 90.7). The fold test is negative at the 99% probability level according to the fold tests of *McElhinny* [1964] and *McFadden* [1990], indicating that the component was acquired after folding. According to the International Geomagnetic Reference Field [*International Association of Geomagnetism and Aeronomy*, *Working group V-MOD*, 2010],

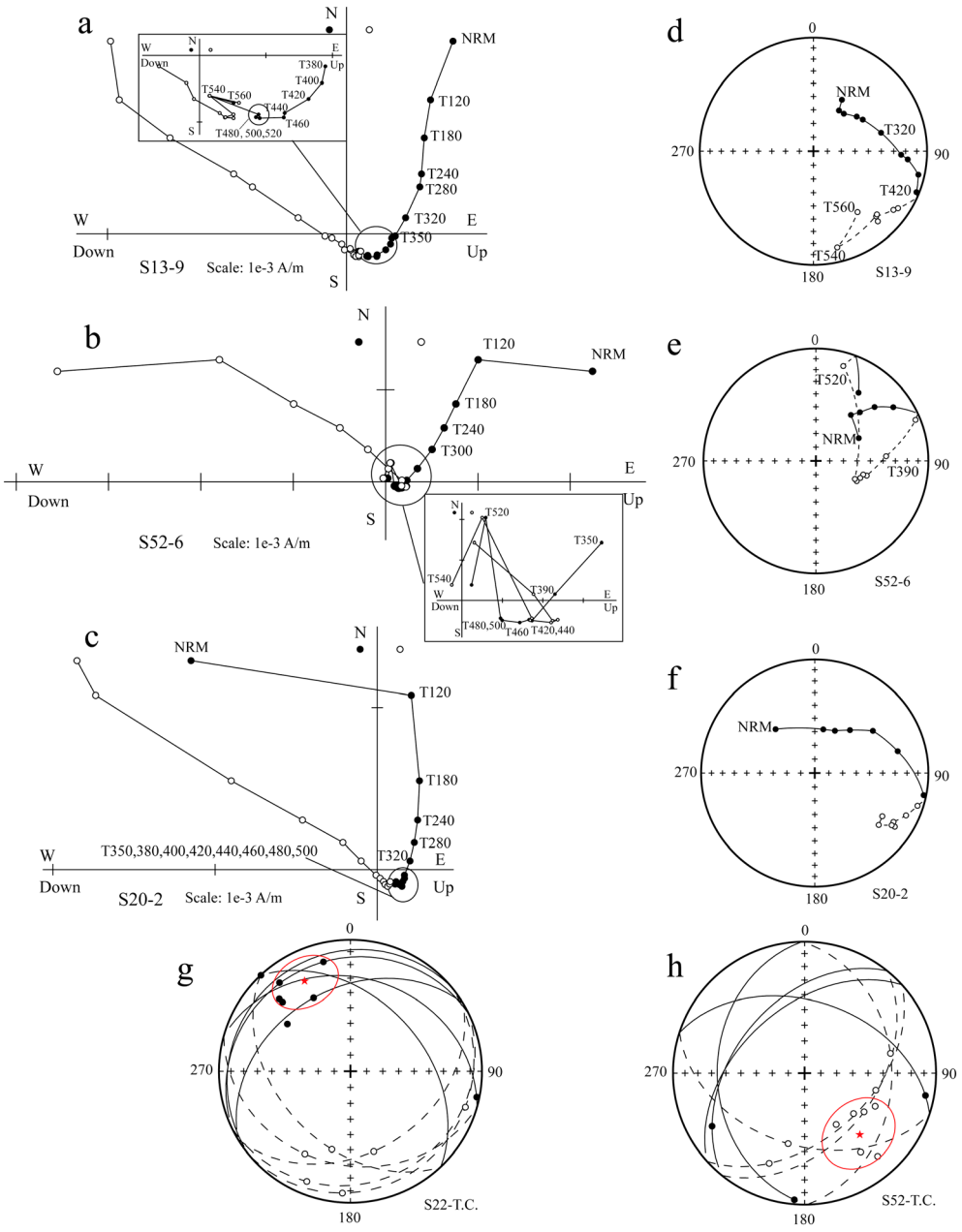


Figure 4. (a-c) Orthogonal vector diagrams illustrating the response of representative specimens from limestones of the Upper Ordovician Pagoda Formation to thermal demagnetization (in situ coordinates). Note the erratic directional changes after heating to 460–540°C. (d-f) Stereographic projections of the change in NRM directions along remagnetization great circles (in situ coordinates). (g and h) Site-mean directions for sites 22 and 52 calculated from the intersection of the great circles with limiting sector constraints and directly observed directions after tilt corrections (TC).

the present Earth field direction (PEF) is $D/I = 357.4°/49.9°$. In comparison, the direction of the low-temperature component is indistinguishable from the PEF, indicating a probable recent viscous magnetization origin.

The high-temperature component can be determined at 26 out of 45 sites, which is directed toward the southeast (northwest) with upward (downward) inclinations. The directions for each site are shown in Table 1. The site-mean direction of the components is Dg = 137.8°, Ig = −36.3° (kg = 8.7, α_{95} = 10.2°) in in situ coordinates, and Ds = 132.6°, Is = −35.2° (ks = 61.7, α_{95} = 3.6°) after tilt correction (Figure 7).

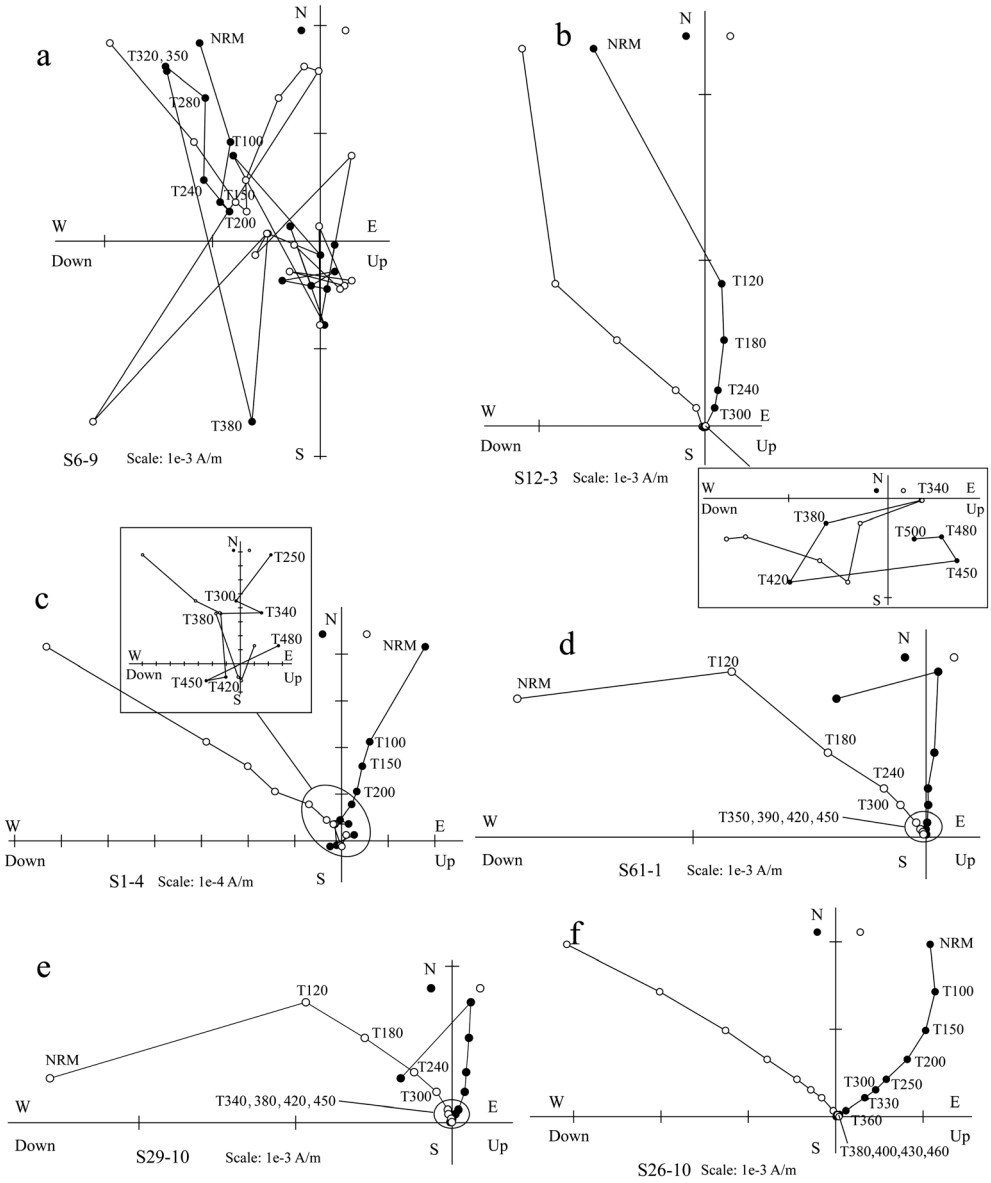


Figure 5. Orthogonal vector diagrams of thermal demagnetization in in situ coordinates for representative specimens from limestones of the Upper Ordovician Pagoda Formation, showing only (a to c) the low-temperature component, (d) a single component close to the present geomagnetic field, (e) a low-temperature component below 300°C and a component directed to NNE from 300°C to 450°C, and (f) arced trajectory below 360°C. Solid (open) symbols denote the projection on the horizontal (vertical) coordinates.

5. Discussion

5.1. Reliability Tests of the Remanent Magnetization

The site-mean directions of the characteristic remanent magnetization are shown in Figure 7 together with the distribution of both normal and reversed polarities. The directions of normal polarity (18 sites) and reversed polarity (8 sites) are antipodal, passing the reversal test of *McFadden and McElhinny* [1990] at the 95% probability level (B class, angle between normal, and reversed mean directions = 7.2°, critical Gamma = 7.6°, and critical *R* at 95% = 0.132947). In addition, a succession of at least seven normal and reversed magnetic intervals is present in two continuous sections of about 11–16 m in thickness, e.g., sites S13–S20 and sites S49–S52 from lower to upper sections.

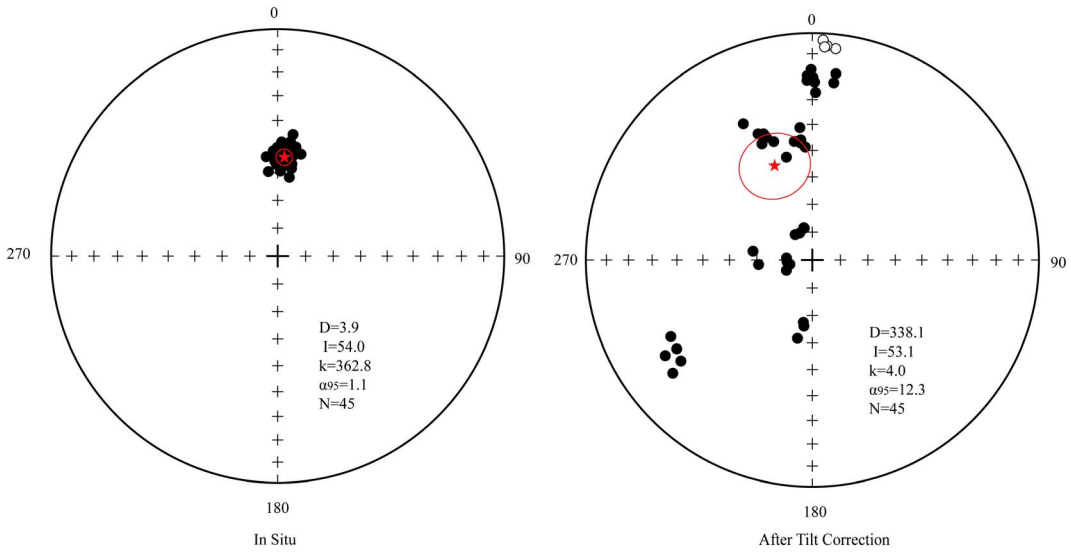


Figure 6. Equal-area stereographic projection of site-mean directions of the low-temperature component in in situ and after-tilt correction. Closed (open) symbols represent lower (upper) hemisphere projection.

The sampling sites are distributed along several flanks of the folds, enabling an effective fold test in several respects. First, we conducted a *sensu strictu* fold test, including sites 3–5, 56, 57, 61, and 62 from the northern limb of the anticline in the southern area; and sites 58–60 from the southern limb. Second, the fold test is designed with the northern limb of a fold (sites 3–5, 56, 57, 61, 62; dip direction to the north) and a northern limb of another fold (sites 13–18, 20, 28, 29; dip direction to the south). Finally, these three limbs from the two folds were taken into account for a tilt test in order to compare the consistency of the high-temperature remanent directions. All three fold tests are positive at the 95% or 99% probability levels according to the fold tests of *McElhinny* [1964] and *McFadden* [1990]. The statistical parameters for the fold tests are summarized in Table 2. Therefore, the high-temperature component was acquired prior to folding which occurred in the Jurassic-Cretaceous [*Wei et al*., 1997; *Pei et al*., 2009; *Sun*, 2011]. The magnetization was acquired at least prior to the Jurassic. We calculated the paleomagnetic pole using the high-temperature component direction (Ds/Is = 132.6°/−35.2°), and compared it with the Phanerozoic apparent pole wander path (APWP) of the Yangtze Block summarized by *Huang et al*. [2008]. The site-mean direction obtained from the high-temperature component yields a paleomagnetic pole at 45.8°S, 191.3°E (dp = 2.4°, dm = 4.2°). In comparison, it is clear that the Late Ordovician pole in this study differs from the available paleomagnetic poles since the Silurian of the SCB (Figure 8). This implies that the high-temperature component is probably a primary remanent magnetization.

5.2. Ordovician Paleomagnetic Poles From the Yangtze Block

There are only two sets of Ordovician paleomagnetic results reported from the Yangtze Block. *Fang et al*. [1990] first reported paleomagnetic results from the Early Ordovician red beds near Kunming City, Yunnan Province, in the southwestern Yangtze Block. In only 5 out of 10 sites from the Hongshiya formation, a total of 29 samples, was it possible to isolate a single magnetic component with a single polarity. Although a fold test was positive, the folding process occurred during the Tertiary. It can only be concluded that the remanence was acquired prior to the Tertiary. It is notable that the paleolatitude calculated from the remanent magnetization data for the Kunming area is 48°S [*Fang et al*., 1990], which is inconsistent with the geological information—as pointed out by *Nie* [1991]. In addition, *Wu et al*. [1999] also published an Early-Middle Ordovician paleomagnetic result from the Xingshan-Zigui section in Hubei Province. One Middle Ordovician site with 13 samples, and three Early Ordovician sites with 13 samples, yielded a site-mean direction of $D/I = 146.2°/17.6°$ ($\alpha_{95} = 13.7°$) after tilt correction. This result was regarded as very preliminary in nature, because no fold and/or reversal tests were applied.

In summary, two previous results based on a limited number of samples provide contradictory paleogeographic interpretations for the SCB, and they are also inconsistent with our new data. Our high-quality Late Ordovician (456 Ma) pole fulfills all of the quality indices of *Van der Voo* [1990].

Table 1. The Site-Mean Directions of the High-Temperature Component From the Upper Ordovician Pagoda Formation in the Wangcang Area[a]

	Dip Direction								
Site	Strike	Dip	*N*(*n*,*c*)	Dg (deg)	Ig (deg)	Ds (deg)	Is (deg)	*k*	α_{95}
S3	263	36	4(4/0)	86.2	−40.1	111.4	−29.7	59.1	12.1
S4–S5	263	36	5(2/3)	116.7	−68.4	150.0	−39.9	30.8	15.9
S10	124	99	5(4/1)	150.9	25.5	148.2	−28.0	56.8	10.5
S13	105	23	10(5/5)	153.4	−33.1	134.6	−48.1	19.5	11.5
S14	105	23	8(6/2)	136.3	−27.3	123.0	−37.1	22.4	12.2
S15	105	23	8(6/2)	139.6	−28.1	124.7	−39.0	24.9	11.5
S16	108	29	11(10/1)	322.4	23.8	306.7	37.1	24.2	9.5
S17	108	29	10(3/7)	146.4	−20.6	126.3	−38.8	22.2	11.0
S18	108	29	7(2/5)	160.3	−18.6	142.1	−45.0	39.5	10.7
S20	109	29	10(7/3)	146.3	−14.8	129.2	−36.1	10.0	16.3
S28–S29	109	29	4(2/2)	165.4	−18.6	143.5	−42.9	28.4	20.3
S21	207	25	12(5/7)	157.7	−55.8	143.5	−34.7	16.6	11.2
S22	207	25	9(4/5)	321.5	43.9	315.6	20.6	26.7	10.6
S23	207	25	7(6/1)	135.3	−60.1	128.1	−35.9	45.9	9.1
S24	207	25	7(5/2)	309.3	57.4	304.8	32.8	135.4	5.3
S25	207	25	6(5/1)	331.9	53.0	320.7	31.0	62.0	8.7
S49	198	14	11(10/1)	320.1	40.9	315.3	28.7	21.9	10.0
S50	198	14	9(7/2)	133.1	−46.2	128.6	−33.3	35.9	8.8
S52	198	14	9(5/4)	143.1	−49.8	136.0	−37.8	22.7	11.4
S53	198	14	11(9/2)	327.9	50.6	319.7	39.2	17.4	11.3
S54–S55	198	14	9(7/2)	317.8	39.9	313.4	27.5	29.4	9.8
S56–S57	274	34	11(8/3)	305.1	52.9	327.8	29.1	18.3	11.1
S58–S59	92	59	10(8/2)	144.5	−2.0	123.9	−44.2	27.6	9.5
S60	92	59	7(4/3)	140.4	9.5	130.5	−33.1	9.7	21.1
S61	276	55	5(2/3)	79.4	−37.7	122.2	−32.4	21.5	19.1
S62	276	55	9(6/3)	87.9	−33.5	120.9	−24.4	9.3	18.0
Mean				137.8	−36.3	132.6	−35.2	61.7	3.6

[a]*N*: number of samples (*n* = number of direct observations; *c* = number of remagnetization great circles); Dg, Ig and Ds, Is : declination and inclination in geographic and stratigraphic coordinates, respectively; *k*: Fisher precision parameters; α_{95}: radius of the 95% confidence circle about the mean direction.

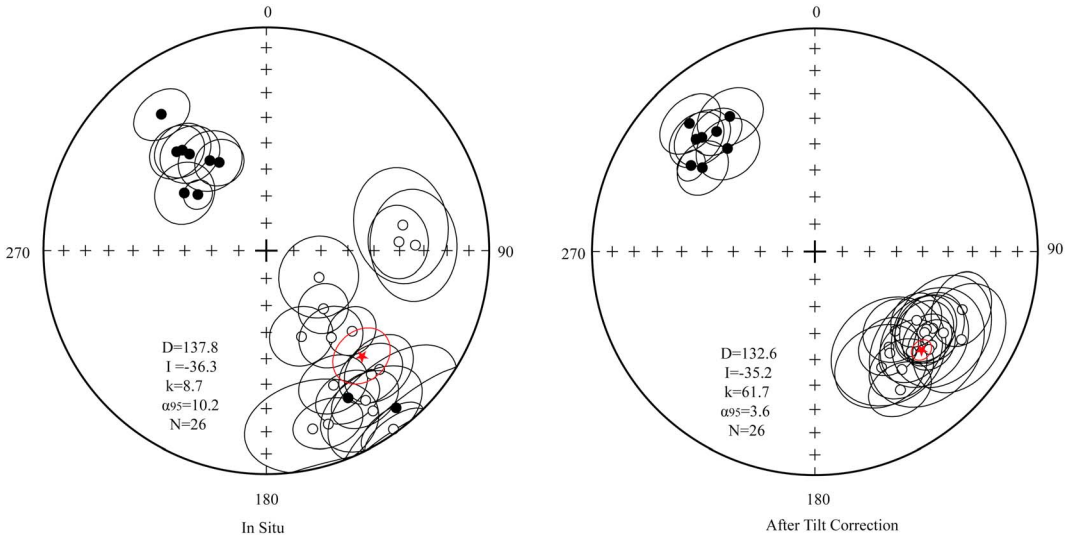


Figure 7. Equal-area stereographic projection of site-mean directions of the high-temperature component in in situ and after-tilt correction. Closed (open) symbols represent lower (upper) hemisphere projection.

Table 2. The Statistical Parameters for Three Types of Fold Test[a]

	McElhinny Method			McFadden Method				
		F(2*(*n*2-1), 2*(*n*1-1))		Critic Xi				
Type	ks/kg	at 5%	at 1%	at 95%	at 99%	Xi2 IS	Xi2 TC	Fold Test
Sensu strictu fold test	6.58057	2.6900	4.1600	3.086	4.253	4.403	0.5964	positive
Two-limbs fold test	5.83721	1.9800	2.6600	6.576	7.331	8.206	2.334	positive
Tilt test	7.14082	1.8833	2.4666	4.510	6.305	9.841	2.504	positive

[a]kg, ks: Fisher precision parameters before and after tilt correction, respectively.

5.3. Paleogeographic Implications

The paleolatitude calculated from magnetic inclination data remains unclear in terms of its hemisphere, especially for the interval prior to the early Paleozoic. The minimum poleward transport between the Silurian pole and the Late Ordovician pole (this study) in Figure 8 is used to constrain the polarity of the magnetic remanence. Thus, the SE upward remanent magnetic direction was of normal polarity. It is noteworthy that the sampling area of the SCB was located in tropical latitudes of the southern hemisphere (19.4°S) during the early Late Ordovician.

Various methods have been used to constrain the paleoposition of the SCB in the early Paleozoic, including geochronology, paleobiogeography, sedimentary facies analysis, and paleomagnetism. However, it remains high debatable whether or not South China was an integral part of Gondwana during the early Paleozoic [*Torsvik and Cocks*, 2009; *Cocks and Torsvik*, 2013; *Burrett et al*., 2014], and the relationship between the SCB and Gondwana is unclear [*Yang et al*., 2004; *Torsvik et al*., 2009; *Cawood et al*., 2013; *Li et al*., 2013]. Based on different paleomagnetic data sets, South China is commonly placed in two different positions relative to Gondwana, either adjacent to Australia [*Yang et al*., 2004; *Cawood et al*., 2013] or India/Pakistan [*Cocks and Torsvik*, 2002; *Torsvik et al*., 2009].

The early Paleozoic paleomagnetic data are sparse; however, faunal province analysis has also been used to constrain the paleoposition of the blocks. *Zhou and Zhen* [2008] performed a cladistic analysis of all of the trilobite genera for China during the Ordovician, and demonstrated that North China, South China, and the northeastern margin of Gondwana were closely related to each other, forming a single faunal province, namely, the peri-Gondwana Realm. The peri-Gondwanan province was further divided into two contrasting subprovinces, one consisting of South China, Tarim, and Indochina, and the other including the North China, Sibumasu, and the Lhasa blocks, during the Middle Ordovician period from the Floian to the early Katian [*Zhou et al*., 2009]. However, *Fortey* [1997] considered that the trilobites in South China were essentially identical to those in Sibumasu in the early Late Ordovician (Sandbian) at about 460 Ma. Based on the Late Ordovician (Sandbian and Early Katian) brachiopod faunas, *Cocks and Zhan* [1998] also suggested that the diverse faunas from South China did not have a high proportion of genera in common with North China, the Chu-Ili Terrane of Kazakhstan or the Australian faunas, and that the Sibumasu part of Burma (Myanmar) and South China had the closest affinity.

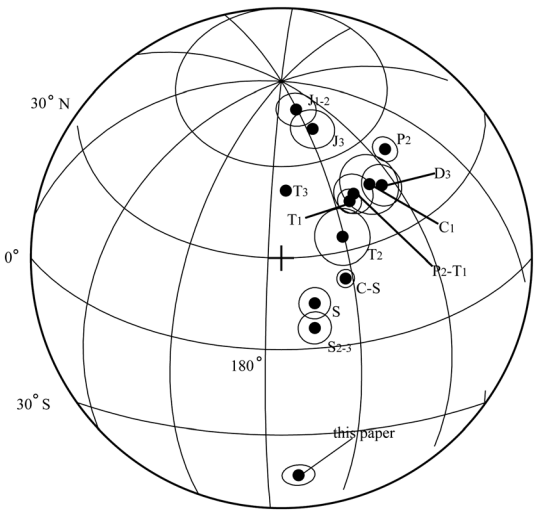


Figure 8. Comparison of paleomagnetic poles obtained in this study and those for the Silurian-Jurassic of the Yangtze Block reported by *Huang et al*. [2008]. S, Silurian; S_{2-3}, Middle to Late Silurian; C-S, Carboniferous to Silurian; C_1, Early Carboniferous; D_3, Late Devonian; P_2, Late Permian; P_2-T_1, Late Permian to Early Triassic; T_1, Early Triassic; T_2, Middle Triassic; T_3, Late Triassic; J_{1-2}, Early to Middle Jurassic; and J_3, Late Jurassic.

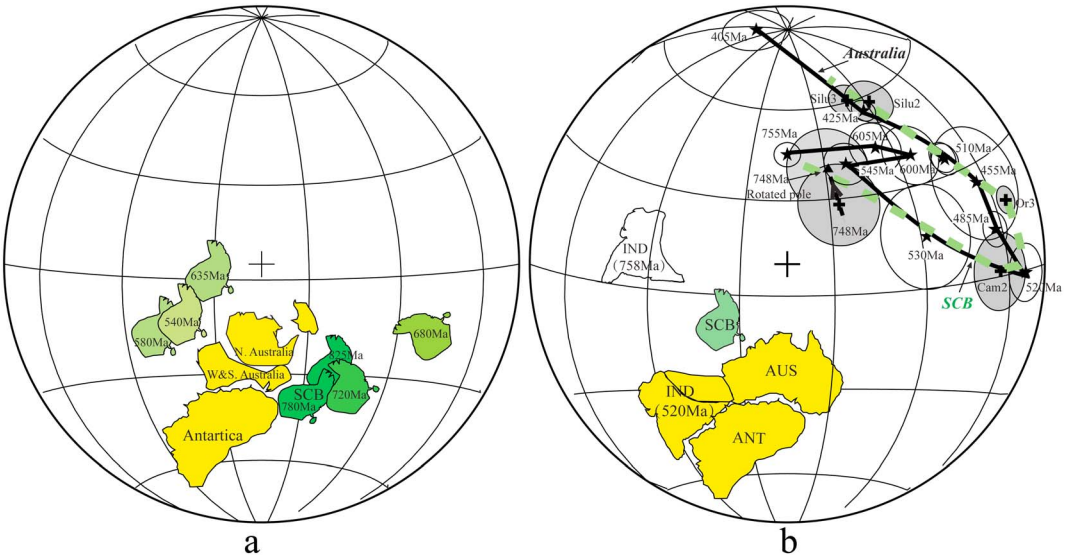


Figure 9. (a) Simplified reconstruction of South China in the present Australian coordinates during the periods of 825–540 Ma, according to the reconstructions proposed by *Li et al.* [2013]. The North Australian craton was rotated with respect to the South and West Australian cratons around an Euler pole at −20°N, 135°E with angle −40° during the late Neoproterozoic (650–550 Ma) [*Li and Evans*, 2011]; (b) Reconstruction of South China and Australia for the mid-Neoproterozoic and early Paleozoic in modern Australian coordinates proposed by *Yang et al.* [2004]. The SCB is rotated around an Euler pole at 10.4°N, 140°E with an angle of 71°. Note that the Late Ordovician pole (this study) falls within the coeval pole of Gondwana. Abbreviations: SCB, South China Block; AUS, Australia; N. Australia, North Australia; W&S. Australia, South and West Australia; IND, India; ANT, Antarctica; Cam2, Middle Cambrian; O_r3, Late Ordovician; Silu2, Middle Silurian; and Silu3, Late Silurian.

Recently, the provenance of siliciclastic strata has been constrained using U-Pb analyses of detrital zircon grains, and this may provide useful information on the separation and/or collision of different blocks. In order to address the issue of Gondwana affinity, an extensive program of the dating of detrital zircon grains separated from late Proterozoic and early Paleozoic rocks in the SCB has been conducted over the last decade. The presence of end Mesoproterozoic and Neoproterozoic peaks in the age spectra of zircons is a characteristic feature of detritus derived from East Gondwana sources—a similar characteristic of the Tethyan Himalaya and younger Paleozoic successions from Western Australia [*Cawood et al.*, 2013; *Xu et al.*, 2013; *Burrett et al.*, 2014]. A common source implies that a close geographic link between South China, Western Australia, and the Tethyan Himalaya existed during the Neoproterozoic and early Paleozoic. However, the orientation of the SCB and the paleoposition of the SCB remain undetermined along the northern margin of East Gondwana [*Yu et al.*, 2008; *Wang et al.*, 2010; *Duan et al.*, 2011; *Xu et al.*, 2013; *Burrett et al.*, 2014].

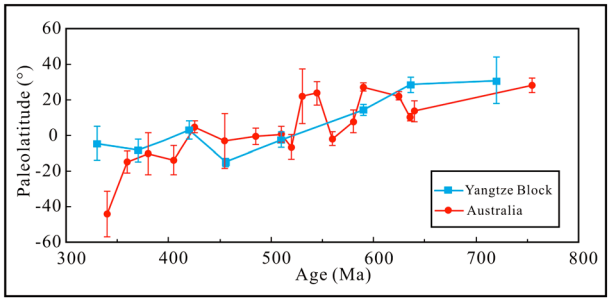


Figure 10. Comparison of paleolatitudes for the Yangtze Block and the Australia from 750 Ma to the late Paleozoic for reference points at 23°N, 103°E (Yangtze Block) and −15°N, 120°E (Australia). The poles yielding the paleolatitudes calculated for the reference point of the Yangtze Block are from *Evans et al.* [2000], *Huang et al.* [2000], *Yang et al.* [2004], *Huang et al.* [2008], *Zhang et al.* [2013, 2014], and this study. The poles yielding the paleolatitudes calculated for the reference point of Australia are from *Wingate and Giddings* [2000], *McElhinny et al.* [2003], and *Schmidt* [2014].

Reconstructions based on paleomagnetism are also varied. *Yang et al.* [2004] suggested a long-standing connection between South China and NW Australia during the mid-Neoproterozoic and early Paleozoic (Figure 9b). However, *Li et al.* [2008, 2013] argued that the SCB was situated in a "missing link" between Australia and Laurentia for the 900–800 Ma interval and then broke away from the middle of Australia and Laurentia during the interval 820–750 Ma, and migrated from northeast Australia to northwestern Australia during

the late Neoproterozoic (Figure 9a). *Zhang et al*. [2013] also proposed that the SCB was situated in the West (at around 750 Ma) and Northwest of Australia (at 635 Ma) and then drifted in a southeastwards direction to collide with Australia prior to the Middle Cambrian. However, evidence for the collisional event that may have led to the juxtaposition of the SCB close to northwest of Australia is clearly absent in the southeast SCB for the interval 635–520 Ma.

Although the late Neoproterozoic reconstruction between the SCB and Australia is still uncertain due to the fact that coeval pair poles are difficult to obtain, e.g., the Liantuo pole and the 755 Ma pole of Australia. However, we note that our new Late Ordovician pole falls within the coeval pole of Gondwana after rotating around an Euler pole at 10.4°N, 140°E with an angle of 71°, implying that South China was situated to the northwest Australia during the early Paleozoic (Figure 9b).

Figure 10 illustrates changes in the paleolatitude of the Yangtze Block and Northwest Australia from 750 Ma to the late Paleozoic at the reference points of 23°N, 103°E (Yangtze Block) and −15°N, 120°E (Australia) according to the poles of the late Proterozoic [*Evans et al*., 2000; *Wingate and Giddings*, 2000; *Zhang et al*., 2013, 2014; *Schmidt*, 2014] and the Paleozoic [*Huang et al*., 2000; *McElhinny et al*., 2003; *Yang et al*., 2004; *Huang et al*., 2008; this study] of both blocks. It is obvious that the paleolatitude changes of both blocks are approximately synchronous during the Middle Cambiran to Middle Devonian, supporting a close linkage of these two blocks in the early Paleozoic [*Yang et al*., 2004].

Some uncertainties remain regarding the paleolatitude change of South China between the Early and early Late Ordovician [*Fang et al*., 1990; this study], which may imply a large motion of about 2000 km from 480 Ma to 456 Ma. Thus, additional reliable Early Ordovician paleomagnetic data are urgently needed to verify the paleoposition of the SCB at this time.

6. Conclusions

A paleomagnetic study of Late Ordovician limestones in the northern Yangtze Block has yielded a high-temperature component likely carried by magnetite. The direction is $D/I = 132.6°/-35.2°$ ($\alpha_{95} = 3.6°$) after bedding correction. It passes both the fold test and the reversal test. Although the folding is of Mesozoic age, the corresponding paleomagnetic pole is significantly different from those for the Silurian to the Triassic obtained from the Yangtze Block. These results suggest that the high-temperature component is probably primary. The high-temperature component gives a paleolatitude of 19.5°S, suggesting that the Yangtze Block was at tropic latitudes during the Late Ordovician. This new paleomagnetic result bridges the Ordovician data gap and further confirms the close proximity of the South China Block and Australia during the Ordovician.

Acknowledgments

Data supporting Figure 6 are available in Table S1 in the supporting information. Data supporting Figure 10 are available in Table S2 in the supporting information. This work was funded by the Chinese National Natural Science Foundation (grant 41230208). Paleomagnetic data were analyzed using J.P. Cogné's (2003) and R. Enkin's computer program packets. We are particularly grateful to Renbin Zhan for guidance in the field and for his help in gathering material. We thank R. Van der Voo and two anonymous reviewers for their comments on the manuscript.

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